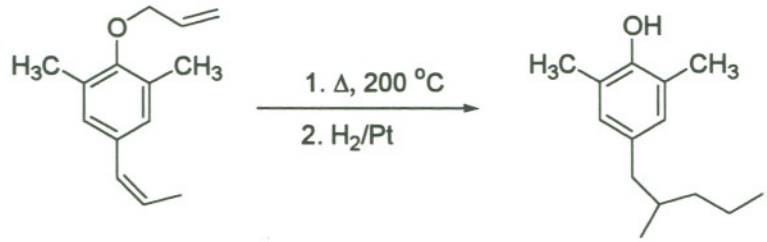
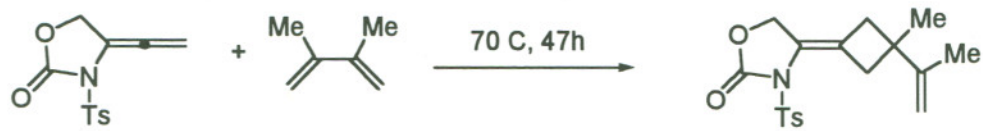
Organic Cume, December 6, 1997

1. Analyze the electrocyclic ring closure of the following anion using a symmetry correlation diagram. Include in your analysis prediction of the stereochemistry of the product.

2. Give the mechanism of the following thermal conversion (the 200 °C step) in terms of allowed reactions. (Nickon, A.; Aaronoff, B.R. *J. Org. Chem.* **1964**, *29*, 3014)



3. In a recent paper the following reaction was found to have excellent regio- and stereoselectivities. It was described as following a Woodward Hoffmann allowed [π2s+(π2s+π2s)] process. Draw appropriate diagrams indicating the orbital interactions consistent with this analysis. (Kimura, M. Horino, Y., Wakamiya, Y., Okajima, T., Tamaru, Y. *J. Am. Chem. Soc.* **1997**, *119(44)*, 10869-10870)